

# Order of Operations with Radicals

## Lesson Plan

### 1. Lesson Information

|        |                                              |              |                                            |
|--------|----------------------------------------------|--------------|--------------------------------------------|
| Course | Grade 7 Accelerated<br>(1205050)             | Unit         | 4 — Solving Problems with Rational Numbers |
| Lesson | Lesson 6 — Order of Operations with Radicals | Date         | _____                                      |
| Length | 53 minutes (one period)                      | Benchmark(s) | MA.8.NSO.1.7                               |

### 2. Standards

**MA.8.NSO.1.7** Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

**Must.** Evaluate multi-step expressions with rational numbers, exponents and radicals, in six steps or fewer.

**Must not.** No mnemonic. The state guide says to avoid PEMDAS because it leaves out grouping symbols, and the word appears nowhere in this unit.

### 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR             | What it asks                                                                        | The evidence in this lesson                                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| MA.K12.MTR. 5.1 | Use patterns and structure to help understand and connect mathematical concepts.    | Notes I — the four levels are read off the expression's structure and the class is never given a mnemonic, because the guide asks us not to give one. |
| MA.K12.MTR. 3.1 | Complete tasks with mathematical fluency.                                           | Boards 1, 3 and 7 — one structural decision each, written and up on the cue; these are the errors that cost marks everywhere else.                    |
| MA.K12.MTR. 4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | Board 5 — when the room splits between 50 and 80, the sixty-second re-vote before the reveal.                                                         |

### 4. Learning Target and Essential Question

**Learning target.** I can evaluate a multi-step expression with rational numbers, exponents and radicals by reading its structure.

**Essential question.** What decides the order in which an expression is evaluated?

**Building on.** 4.05 — the radical bar as a grouping symbol; 3.04–3.05 — negative exponents on integer and rational bases; Grade 6 — the order of operations with whole numbers.

**Working toward.** 4.07–08 — the same structure inside real-world problems; MA.912.NSO.1.4 in Algebra 1.

## 5. Sequence

*read from the deck; never typed*

| Segment                             | Min       | What the teacher does                                                                        | What students do                           |
|-------------------------------------|-----------|----------------------------------------------------------------------------------------------|--------------------------------------------|
| Settle in; post the learning target | 1         | Post the target; say the 'today' line and nothing else yet.                                  | Copy the target.                           |
| Warm-Up                             | 5         | Two minutes silent, then reveal. One sentence of reteach per question at most.               | Four retrieval questions, alone, no notes. |
| Notes                               | 12        | Teach from the front. Expand once per law; say the OFF line out loud.                        | Copy the notes; answer the checks aloud.   |
| Example 1                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Example 2                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Whiteboards                         | 12        | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three.    |
| Independent Set                     | 6         | Silent. Circulate and mark what you see; do not teach.                                       | Six questions, written, alone.             |
| IXL                                 | 5         | Assign the day's IXL skills.                                                                 | Start the skills; finish tonight.          |
| <b>Total</b>                        | <b>53</b> |                                                                                              |                                            |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question                                                                                                                                                         | DOK   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Why is $-2^2$ different from $(-2)^2$ ? What is the exponent attached to in each one?                                                                            | DOK 3 |
| A mnemonic for the order of operations has no letter for the radical bar or the fraction bar. What does reading the structure give you that the mnemonic cannot? | DOK 3 |
| Roscoe says a square is always positive. What would have to be true about $\sqrt{n} - 11$ for that to be right, and what value of $n$ breaks it?                 | DOK 3 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks                                                                                                                             | Answer                                             | If the board is wrong                                                                                                                                                                                                                                                                                                     |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $12 \div 3 \cdot 2$                                                                                                                             | <b>8</b>                                           | $2 - 3 \cdot 2$ was done first [8.NSO.1.7: multiplication and division are one level, taken left to right]; $72 -$ the division was read as a multiplication                                                                                                                                                              |
| 2 | A square tile has side $\sqrt{196}$ centimetres. Its diagonal is about 19.8 centimetres. What is the perimeter of the tile?                     | <b>56 centimetres</b>                              | anything built from 19.8 — the unneeded figure was used [a perimeter needs the side]; 14 — the side, not the perimeter; 196 — the root was never taken                                                                                                                                                                    |
| 3 | $10 - 4 + 3$                                                                                                                                    | <b>9</b>                                           | $3 - 4 + 3$ was done first [8.NSO.1.7: addition and subtraction are one level, taken left to right]                                                                                                                                                                                                                       |
| 4 | $\frac{2^3 - 5}{6^2 - 27}$                                                                                                                      | <b>1/3</b>                                         | anything that divides before the top and bottom are finished [8.NSO.1.7: the fraction bar groups both]; $1/12 - 6^2$ read as $6 \cdot 2$                                                                                                                                                                                  |
| 5 | $2 + 3 \cdot 4^2$ . Which is the value?                                                                                                         | <b>A — 50</b>                                      | (B) $2 + 3$ was done first, then $5 \cdot 16$ [8.NSO.1.7: multiplication comes before addition]; (C) the exponent was applied to $3 \cdot 4$ instead of to the 4 alone [8.NSO.1.7: an exponent belongs to its own base]; (D) $4^2$ was read as $4 \cdot 2$ [8.NSO.1.3: an exponent is not a factor]                       |
| 6 | $(6^2 - 4)$ over $2^3$ . Which is the value?                                                                                                    | <b>A — 4</b>                                       | (B) only the 4 was divided by $2^3$ before the subtraction [8.NSO.1.7: the bar groups the whole numerator]; (C) $6^2$ was divided by $2^3$ first and then 4 was subtracted [8.NSO.1.7: the bar groups the whole numerator]; (D) the numerator was multiplied by $2^3$ instead of divided [8.NSO.1.7: the bar is division] |
| 7 | $-2^2 + (-2)^2$                                                                                                                                 | <b>0</b>                                           | $8 -$ both terms read as $(-2)^2$ [8.NSO.1.7: an exponent attaches to its own base, and parentheses decide what that base is]; $-8 -$ both terms read as $-2^2$                                                                                                                                                           |
| 8 | $12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$                                                                                            | <b>17/6</b>                                        | $12/8 + 12/9$ left unadded — a value, not an answer; $1/8 + 1/9 = 2/17 -$ the denominators were added [Grade 5: adding fractions needs a common denominator]                                                                                                                                                              |
| 9 | Roscoe says that $(\sqrt{n} - 11)^2$ is always positive, because a square is always positive. Is he right? Give a value of $n$ that settles it. | <b>No — at <math>n = 121</math> the value is 0</b> | 'yes, squares are positive' — the claim repeated rather than tested [8.NSO.1.7: a square is never negative, which is a different statement]; 'no' with no value of $n -$ the counterexample is the evidence                                                                                                               |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / IEP.** Give the four-level table from Notes I as a card and let it stay out. On every expression the student first circles all the grouping symbols and numbers them from the inside out, and only then evaluates; the circling is the work, and the arithmetic follows it.

**ELL.** Grouping symbol is the phrase to own. Put the four of them on the board — parentheses, brackets, the radical bar, the fraction bar — and have the student point to each one in an expression and name it before evaluating anything.

**Enrichment.** Ask for an expression using 2, 3 and 4 once each, with any grouping symbols, whose value is as large as possible — then as small as possible. Then ask why a mnemonic could not have answered either question.

## 10. Closure, Vocabulary, Materials, Practice

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**Closure.** Board 9 is written work and it is the exit evidence: the student has to give a value of  $n$  that settles Roscoe's claim and say what is wrong with the sentence, not only that it is wrong. Read the boards, not the papers.

**Vocabulary.** **grouping symbol; nested; multi-step expression**

**Materials.** Whiteboards and markers. Calculators are allowed on everything, and this is the lesson where a calculator is worth using against the class: enter  $-2^2$  and  $(-2)^2$  and let the machine settle it.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Evaluate radical expressions — D8J.